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Business Problem

The following analysis aims to extract **three main actionable insights** from flight booking data:

1.Flight Clustering

Identify meaningful groups of flights based on their characteristics and understand what differentiates them.

2.Pricing Strategy Evaluation

Compare pricing strategies **A and B** using relevant performance metrics and determine which strategy performs more effectively.

3.Flight Classification

Develop a model capable of predicting the group (**UKP or CANARY**) of a flight and assess how reliably this classification can be performed.

Data Overview

The dataset contains information on **flight departures, bookings, passengers, fares, and flight activity**.

	Variables
	Flight departure date
	Pricing strategy/model used
	Flight identifier
	Booking timestamp
	Passenger identifier
	Fare paid in euros
	Indicator of whether the booking/flight record is active
	Flight route category (classification task)

1. Flight Clustering

Approach

To understand the different types of flights in the dataset, the analysis was conducted at **flight level**, rather than at individual booking level.

For each flight, a set of indicators was created to capture its main commercial and operational characteristics:

- **Booking volume**: number of bookings
- **Average fare**: typical fare paid for the flight
- **Cancellation rate**: proportion of bookings that were cancelled
- **Weekend indicator**: whether the flight departed on a weekend
- **Month**: period of the year in which the flight operated

Flights were then grouped according to similarities across these characteristics. The number of groups was selected based on the separation between groups, with **6 clusters** providing a useful and interpretable segmentation.

Cluster Findings

The analysis identified **six distinct flight profiles**:

Cluster	Profile	Main Characteristics
0	High segment - Weekend	Weekend departures in late July, combining seasonal and weekend effects
1	Mid segment - Weekend	June weekend flights with moderately higher fares than weekday equivalents
2	Premium Segment - Weekday	Weekday flights with distinctly higher fares despite normal booking volumes and cancellation rates
3	Weekday high volumes	Late July weekday flights with the highest booking volumes and low fares
4	Low segment - Weekday	June weekday flights with low fares, standard booking volumes and cancellation rates
5	Cancelled / Disrupted Flights	Flights characterised by a near-total cancellation rate

Business Interpretation

- The clustering reveals that **timing, fare level and operational reliability are key dimensions differentiating flights**.
- Early-summer flights are generally associated with lower fares, whereas late-July weekends command higher fares.
- The analysis also identifies a distinct **premium segment**, suggesting that some flights achieve substantially higher fares without showing a corresponding increase in booking volume or

cancellation behaviour.

- The **disrupted-flight cluster** stands apart from the other segments because of its exceptionally high cancellation rate. This group should be considered separately when interpreting commercial performance, as operational disruption can strongly influence both bookings and fares.
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Opportunities for Improvement

The current segmentation provides a useful first view, but it could be enriched with additional flight-level information.

Potential future features include:

- **Days between booking and departure**, to capture booking behaviour and demand timing
- **Load factor or capacity utilisation**, to distinguish high-demand flights from flights with simply high booking counts
- **Departure time**, to identify morning, afternoon and evening demand patterns
- **Historical fare levels**, to identify unusually expensive or discounted flights
- **Route or airport information**

Potential Production Use Case

Once validated on new data, the clustering model could be used as an **automated flight segmentation tool**.

An effective flight segmentation could:

1. **Support pricing decisions** by comparing a flight with similar historical flights.
2. **Identify operational risks**, particularly flights showing characteristics associated with disruption.
3. **Compute the potential revenue of a flight**, and compare the actual performance to such benchmark
4. **Trigger targeted actions**, such as closer monitoring of high-demand flights or premium-fare opportunities.

Segmentation strategy should be periodically reviewed and recalibrated as **seasonality, customer behaviour and pricing patterns change**. New clusters should be evaluated against the existing segmentation to ensure that the business interpretation remains meaningful.

2. Pricing Strategy Evaluation

Approach

The objective of this analysis is to assess whether **pricing strategies A and B lead to different flight-level outcomes**.

The comparison was performed at **flight level**, using four key indicators:

- **Number of bookings**: measures the volume of demand generated by each flight.
- **Cancellation rate**: captures the proportion of bookings that were cancelled.
- **Average fare**: measures the typical fare paid for a flight.
- **Average revenue**: captures the average revenue generated per flight (active flight only)

Both descriptive statistics and statistical tests were used. This allows us to distinguish between **small differences observed in the data** and differences that are sufficiently strong to provide evidence of a systematic effect.

Results

The two strategies produce remarkably similar outcomes across the main business indicators.

Metric	Strategy A	Strategy B	Statistical evidence
Bookings	107.1	106.7	p = 0.009, δ = 0.026
Cancellation rate	4.00%	3.69%	p = 0.865, δ = 0.002
Average fare	€84.16	€83.97	p = 0.716, δ = -0.004
Average revenue	€84.36	€84.01	p = 0.829, δ = -0.002

The only statistically detectable difference concerns **booking volume**. Strategy A has a slightly higher average number of bookings than Strategy B.

However, the effect size is extremely small (**Cliff's δ = 0.026**), indicating that the practical difference between the two strategies is negligible.

For cancellation rate, average fare and average revenue, the statistical tests provide **no evidence of a systematic difference** between the strategies. Their effect sizes are also effectively zero.

Business Interpretation

- Overall, the analysis does **not provide evidence that either pricing strategy produces materially different commercial outcomes**.
- Therefore, based on the available data and the metrics analysed, **there is no clear evidence that changing from one strategy to the other would materially improve performance**.
- Since bookings and revenues are not significant, the decision on which strategy to apply could be taken based on different information (such as customers satisfaction)

Opportunities for Improvement

The current analysis compares the strategies at an aggregated flight level. Further analysis could provide a more precise understanding of when one strategy might perform differently.

Potential improvements include:

- **Control for flight characteristics**, such as route, season, weekday/weekend and departure time.
- Compare performance at different points before departure.
- Investigate whether the strategies behave differently under stricter filters such as **high- and low-demand conditions**.

3. Flight Classification

Approach

The final objective was to predict whether a flight belongs to the **CANARY** or **UKP** group.

Two classification models were initially considered: **Logistic Regression and Random Forest** for the final comparison, while KNN was not retained for the final model selection.

The models were evaluated using several complementary metrics:

- **Accuracy**: overall proportion of correctly classified flights.
- **Precision**: how reliably the model identifies each group.
- **Recall**: how effectively the model identifies flights belonging to each group.
- **F1-score**: balance between precision and recall.

Results

The classification task proved highly predictable, with both retained models achieving strong and consistent performance.

Model	Accuracy	Precision	Recall	F1
Logistic Regression	93.1%	92.8%	92.7%	92.7%
Random Forest	94.3%	94.1%	94.1%	94.0%

Random Forest achieved slightly higher predictive performance, but the difference from Logistic Regression was relatively limited.

The models also showed **substantial overlap in their predictions**, suggesting that they capture similar underlying patterns in the data.

What Drives the Classification?

Both models highlight **fare as the dominant predictor**.

This is consistent with the descriptive analysis: the average fare differs substantially between the two groups:

Route Group	Average Fare
CANARY	€140.29
UKP	€56.64

This large difference makes fare a particularly informative variable for distinguishing the two groups.

For Random Forest, the relative importance of the variables was:

Variable	Importance
Average fare	65.5%
Fare variability	15.2%
Number of bookings	9.7%
Cancellation rate	6.0%
Month	2.3%
Weekend	1.2%

The Logistic Regression model provides a consistent picture, with average fare also having the strongest relationship with the predicted group.

This consistency between the two approaches increases confidence that **fare is the primary signal driving the classification**.

Model Selection

Although Random Forest provides slightly better predictive performance, **Logistic Regression was selected as the final model**.

The decision reflects the relatively small performance difference combined with the advantages of a simpler and more interpretable model.

Logistic Regression makes it easier to understand how the input variables contribute to the classification and provides a more transparent model for business users.

The objective is therefore not simply to maximise predictive performance, but to balance **performance, interpretability and operational simplicity**.

Key Risk: Dependence on Fare

The strong predictive power of fare is both an advantage and a potential production risk.

A model that relies heavily on fare can perform very well when the fare distribution is stable, but its performance may deteriorate if pricing policies change.

Finally, if the model needs to operate before a fare is established, the classification should be rebuilt using variables available at that point in time.

Opportunities for Improvement

The models used provide a strong baseline, but several improvements could be explored before production deployment:

- Perform **hyperparameter tuning**, particularly for Random Forest.
- Test additional classification algorithms and compare their performance.
- Optimise the classification threshold depending on the relative business cost of false positives and false negatives.
- Perform more detailed error analysis to understand which flights are misclassified.

Production Implementation & Maintenance

The final Logistic Regression model can be deployed as a **versioned scikit-learn pipeline**, including all required preprocessing and the classifier.

A production service could receive flight-level data, validate the expected inputs, apply the same preprocessing used during training and return the predicted route group.

The model should be versioned together with its training data definition, code, dependencies and evaluation metrics. Once deployed, its **prediction accuracy and input-data distributions should be monitored over time**.

Retraining should be triggered when performance deteriorates, the underlying data changes significantly, or the business process changes — particularly if pricing practices change and fare is no longer representative of the relationship observed during training.
